

Project 3 – Chessboard Image Generation from FEN

Overview

In **Project 3**, each submission must implement a final evaluation function that receives a **ground-truth chess position (FEN)** and a **viewpoint specification**, and generates:

1. A **synthetic chessboard image**
2. A **realistic chessboard image**
3. A **side-by-side comparison image**

The outputs must be saved to disk in a **fixed directory structure**.

These results will be used for **manual evaluation**, therefore **file paths, formats, and naming conventions must be followed exactly**.

Required Function

Students must implement the following function:

```
def generate_chessboard_image(fen: str, viewpoint: str) -> None:
    """
    Generate synthetic and realistic chessboard images from a given
    FEN.
    """
```

Input Specification

fen

- Type: `str`
- Format: **FEN string**
- Must follow **exactly the same format and conventions** as the ground-truth FENs provided in the CSV files for Projects 1 & 2.

viewpoint

- Type: `str`
- Allowed values:
 - `"white"` – white-side view
 - `"black"` – black-side view

The viewpoint defines the **camera orientation/board orientation** in the generated images, it just means if black pieces or white pieces are closer to the camera.

Output Specification

The function **must save images to disk** under the following directory:

`./results/`

All images must be saved **independently**, not returned from the function.

Required Output Files

For each (`fen`, `viewpoint`) pair, the following files must be generated:

1. Synthetic Image

Path:

`./results/synthetic.png`

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- Description:
 - A clean, synthetic rendering of the chessboard as you used in your data.
 - No photorealism required here (it's the synthetic image)
 - Must accurately reflect the input FEN and viewpoint

2. Realistic Image

Path:

`./results/realistic.png`

- Description:
 - A realistic image derived from the synthetic image
 - Must preserve as best as you can the **board state and viewpoint in the fen and synthetic image you used.**

3. Side-by-Side Comparison Image

Path:

`./results/side_by_side.png`

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- Description:
 - A single image showing:
 - **Left:** synthetic image
 - **Right:** realistic image
 - Images must be aligned and clearly visible

Image Format Requirements

- File format: **PNG**
- Color format: RGB

Resolution is **not fixed**, but all three images must have **compatible dimensions** such that side-by-side visualization is clear.

Viewpoint Consistency

- The generated images must reflect the given `viewpoint`:
 - `"white"` → white pieces closest to the camera
 - `"black"` → black pieces closest to the camera
 - No assumptions about chess notation orientation are allowed beyond the viewpoint input.
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Additional Constraints

-  Do not return images from the function
 -  Do not require interactive input
 -  Do not overwrite files outside `./results/`
 -  The function must be deterministic for the same `(fen, viewpoint)`
 -  The function must create the `./results/` directory if it does not exist
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